



AGU IE 212

DETERMINISTIC OPTIMIZATION

FINAL PROJECT REPORT

Game Balancing



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1.INTRODUCTION

Wide Game Studio, based in Kayseri, one of the largest cities in Turkey, was founded in 2019 and its founders are nine engineering students studying at Abdullah Gül University. They came together to form Wide Game Studio for virtual reality game development. The team initially faced some difficulties in transitioning to the world of virtual reality, but despite this, the members showed determination and completed some project-based work to obtain the funds with which they entered the new world. AGÜ Rector Prof. to the students who came together due to the COVID-19 Pandemic outbreak Dr. with the support of CENGİZ Yılmaz, a place was allocated in the incubation center on the AGÜ campus. The team, which continued its work there for a while, is currently operating in Kayseri Technopark. They developed their game, known as “Flare Frenzy”, in just 4 months. Wide Game Studio, which first started operating in 2018, was established and started to receive investments worth 1.2 dollars. Kültepe Investment contributed to the investment and there are investors from Kayseri as well as angel investors. According to Ramazan Mert Ural, the company's manager, it is aimed to employ 20 people by the end of 2024. Likewise, the company's market value is 1.2 million dollars. Wide Game Studio strives to become a company that brings VR games to the masses, and its primary goal is to create interactive content using the latest achievements and highly advanced content in the field of VR technologies and non-linear storytelling. The staff team created and strengthened is based on providing a better and emotional experience to play and participate in games such as strategic battle simulator, survival, exploration and other game genres related to the creation of Wide Game Studio.



Figure 1. Team of Wide Game

2.THE SYSTEM UNDER CONSIDERATION

In this study, we will focus on game balancing. The project focuses on the game Wizard Legacy, developed by Wide Game Studio. In the play, a father tries to heal his child. For this to happen, the things requested by a book called Hex'O'Cura must be realized. The wish is to cleanse all the evil in the dungeon. There are different types of spells in the game that will kill creatures. Some spells are better as protection, while some spells are stronger as attack. The spells that can be earned differ depending on the levels. Looking at the spells in the attack category, Frost Spike, Lightning Bolt and Flame Ball damage the enemy differently depending on their range.

Defense	Effect
Ward	When the forearms are brought together, a shield is formed.
Wall of Nature	The player creates a wall from the ground. This wall remains active for 10 seconds. This wall has no life, it disappears when its duration expires. It has no combo.
Ground Spikes	When the AD move is performed, thorns appear on the ground that pierce enemies.
Push	When the AD move is made, the enemies in front of the player are sprayed back.

Table 1. Defense and Effect

Spells in the defensive category are Ward, Wall of Nature, Ground Spikes and Push. These spells protect the character from enemies in different ways.

Attack	Effect
Frost Spike	The player scatters short-range ice particles. Deals damage and slows enemies for two seconds. At higher levels, enemies have a chance to freeze completely and remain stationary.
Lightning Bolt	The player throws long-range electricity. Deals damage, bounces to another enemy; The number of bounces increases as the level increases.
Flame Ball	The player launches medium-range fireballs. Deals damage, sets enemies on fire; Burned enemies take damage spread over 2 seconds.

Table 2. Attack and Effect

Ultimate spells provide the character with certain features or prevent him from taking damage for a certain period.

Ultimate	Effect
Overcharge	When the AD move is made, the enemies in front of the player are sprayed back.
Pierce Bow	The player gets a bow and an arrow.
Untouchable	The player takes no damage for a certain number of seconds.

Table 3. Ultimate and Effect

3.METHODOLOGY

3.1. Problem Definition

In virtual games, it is extremely important to ensure in-game balance so that players can have a fair, entertaining and sustainable experience. The meaning of in-game balance is that various characters, equipment, abilities and strategies can compete fairly against each other. However, when this balance is disrupted, some players or strategies may become stronger than others. Thus, it becomes inevitable that one side of the game will constantly be superior and the other side will remain at a disadvantage. Therefore, discontent among players increases, the sense of justice is shaken, and the competitive structure of the game is disrupted.

For example, if a character or piece of equipment is overpowered, other options may be disabled in-game. This situation causes the game to become monotonous and restricts the players' freedom to try various strategies. Likewise, paying players may gain an unfair advantage when microtransactions or in-game purchases lead to uneven power increases.

When all these factors combine, the long-term success of the game and the continuity of the player community is significantly compromised. Therefore, game developers need to constantly monitor and adjust the balance in the game and provide users with a balanced gaming experience. This process involves several stages that include taking into account player feedback, conducting data analysis, and performing ongoing testing. In this way, in-game balance is ensured and players are offered a satisfactory gaming experience.

3.2. Literature Review

The project handles the Game Balancing Problem. In the project of Moffett (2010) aim to balance the game and his goal was to set the objective function as "fun" and maximize it. It has been found that there are many different decision functions and parameters to maximize "fun". The parameters were divided into direct and indirect effects on "fun", and the factors that would directly affect it were determined as follows: labeled player, ambiance, game metrics and

perceived fairness. Just as the focus of the project is, game metrics directly affect the player's enjoyment of the game, and this parameter is important for the development of the model.

Moffett et al. (2010) focused on factors such as score, duration, kill ratio and accuracy in game metrics. When these factors were kept within a certain value range, it was calculated how factors that would affect the player's gaming pleasure, such as playing time, were affected. A model was created with statistical data by determining the degree of correlation between the factors. This study shed light on the fact that deterministic data will be used in the current project.

Another study conducted by Bakkes et al. (2011) focuses on “Player Modelling” for game balancing. Player modelling depends on player’s behaviour in a game environment and to raise the playing strength of a (computer-controlled) player by allowing it to adapt to its (human) opponent and exploit his weaknesses. Also, player modelling considers a player's actions, tactics, strategies, and underlying motivations to understand their overall game behavior. The concept of player modeling had a significant impact on creating constraints in the game balancing project.

As a third, Kına (2023) states that When players must make the best choices possible in scenarios involving strategic interactions, game theory is crucial. Understanding player strategies, figuring out the best strategies, and examining possible outcomes are all part of game theory. Additionally, to better understand game theory, machine learning algorithms can be used to model and understand strategic interactions and player behavior using the analytical tools of game theory. By learning from data, machine learning algorithms create models and forecast outcomes. These algorithms can be developed and optimized using concepts from game theory.

Lastly, optimizing the game balancing problem with mathematical modeling is not a common situation in the literature. However, learning some basic concepts in this field and projects done without using modeling shed light on the creation of the current game balancing project.

3.3. Solution Methodology

The names of spells are defined as a,b,...,h. and each spells have three different levels which get stronger as the level goes up.

Each spell has some features and these features are defined as 1,2,..11.

Here are the values of spells:

1) Spell A : Untouchable Spell

Indices	Features	Level 1	Level 2	Level 3
1	Damage	0	0	0
2	Trigger Cooldown	0	0	0
3	Cooldown	0	0	0
4	Duration	5	7	12
5	Range	0	0	0
6	Projectile speed	0	0	0
7	Projectile Lifetime	0	0	0
8	Effect Duration	0	0	0
9	Usages	0	0	0
10	Prefab	0	0	0
11	Custom stats	0	0	0

Table 4. Spell A: Untouchable Spell

2) Spell B: Earth Wall Spell

Indices	Features	Level 1	Level 2	Level 3
1	Damage	0	0	0
2	Trigger Cooldown	0	0	0
3	Cooldown	3	3	3
4	Duration	15	15	15
5	Range	3	3	3
6	Projectile speed	0	0	0
7	Projectile Lifetime	0	0	0
8	Effect Duration	0	0	0
9	Usages	0	0	0
10	Prefab	0	0	0
11	Custom stats	0	0	0

Table 5. Spell B: Earth Wall Spell

3) Spell C: Ground Spike Spell

Indices	Features	Level 1	Level 2	Level 3
1	Damage	5	7	10
2	Trigger Cooldown	0	0	0
3	Cooldown	0	0	0
4	Duration	0	0	0
5	Range	0	0	0
6	Projectile speed	0	0	0
7	Projectile Lifetime	0	0	0
8	Effect Duration	0	0	0
9	Usages	0	0	0
10	Prefab	1	1	1
11	Custom stats	5	5.5	6

Table 6. Spell C: Ground Spike Spell

4) Spell D: Lighting Bolt Spell

Indices	Features	Level 1	Level 2	Level 3
1	Damage	25	27	30
2	Trigger Cooldown	0.5	0.5	0.5
3	Cooldown	5	5	5
4	Duration	12	15	15
5	Range	15	15	15
6	Projectile speed	0	0	0
7	Projectile Lifetime	0	0	0
8	Effect Duration	0	0	0
9	Usages	0	0	0
10	Prefab	1	1	1
11	Custom stats	1.8	1.9	2.8

Table 7. Spell D: Lighting Bolt Spell

5) Spell E: Bow and Arrow Spell

Indices	Features	Level 1	Level 2	Level 3
1	Damage	20	25	30
2	Trigger Cooldown	0.1	0.1	0.1
3	Cooldown	0	0	0
4	Duration	15	17	18
5	Range	50	60	60
6	Projectile speed	50	50	50
7	Projectile Lifetime	0	0	0
8	Effect Duration	0	0	0
9	Usages	3	4	5
10	Prefab	0	0	0
11	Custom stats	0	0	0

Table 8. Spell E: Bow and Arrow Spell

6) Spell F: Warding Shield Spell

Indices	Features	Level 1	Level 2	Level 3
1	Damage	0	0	0
2	Trigger Cooldown	0	0	0
3	Cooldown	0	0	0
4	Duration	0	0	0
5	Range	0	0	0
6	Projectile speed	0	0	0
7	Projectile Lifetime	0	0	0
8	Effect Duration	0	0	0
9	Usages	0	0	0
10	Prefab	1	1	1
11	Custom stats	70	100	150

Table 9. Spell F: Warding Shield Spell

7) Spell G: Flame Ball Spell

Indices	Features	Level 1	Level 2	Level 3
1	Damage	10	12	15
2	Trigger Cooldown	1	1	1
3	Cooldown	5	5	5
4	Duration	10	12	12
5	Range	500	500	500
6	Projectile speed	10	10	10
7	Projectile Lifetime	15	15	15
8	Effect Duration	0	0	0
9	Usages	0	0	0
10	Prefab	1	1	1
11	Custom stats	2	2	2

Table 10. Spell G: Flame Ball Spell

8) Spell H: Frost Spike Spell

Indices	Features	Level 1	Level 2	Level 3
1	Damage	5	7	10
2	Trigger Cooldown	2	2	1
3	Cooldown	7	5	5
4	Duration	10	10	15
5	Range	500	500	500
6	Projectile speed	20	20	20
7	Projectile Lifetime	15	15	15
8	Effect Duration	0	0	0
9	Usages	0	0	0
10	Prefab	1	1	1
11	Custom stats	5	5	7

Table 11. Spell H: Frost Spike Spell

Power Coefficients of spells' features

Each feature of the spells has certain coefficients which indicate the power or effect of the certain features on the enemy.

The values obtained by developers of the game.

The coefficients can vary depending on the player's experience or skills, here they are represented just averagely.

The values of the features of the particular spells will be multiplied by corresponding power coefficients and the sum of the products will determine the overall effectiveness of the spells.

Indices	Features	Power Coefficients
1	Damage	5
2	Trigger Cooldown	-1
3	Cooldown	-3
4	Duration	1
5	Range	3
6	Projectile speed	3
7	Projectile Lifetime	2
8	Effect Duration	4
9	Usages	1
10	Prefab	1
11	Custom stats	8

Table 12. Power Coefficients of Spells' Features

Values of the enemies:

In the game there are 10 stages and in these 10 stages there are 10 different enemies.

The name of the stage also determines the level of the enemy.

Each enemy has 8 different features, and these features get stronger as the level of the enemies gets higher.

While the first features of enemies is about the health of the enemy, other features specifies the attack skills of the enemies.

Indices	Features	Level 1	Level 2	Level 3	Level 4	Level 5
1	Current health	100	150	200	260	310
2	Damage	0.5	0.6	0.9	1.3	1.5
3	Light Damage	0.5	0.7	1	1.2	1.4
4	Heavy damage	0.5	0.9	1.1	1.3	1.4
5	Malee Attack Range	5	7	10	14	17
6	Ranged Attack Range	7	8	13	15	18
7	Move Speed	3	3.3	6	8	9
8	Attack Cooldown	1.5	1.5	1.4	1.3	1.2

Table 13. Value's of the Enemies 1

Indices	Features	Level 6	Level 7	Level 8	Level 9	Level 10
1	Current health	360	420	490	900	1200
2	Damage	1.7	2	2.3	2,8	3,1
3	Light Damage	1.6	1.9	2.5	3	3,6
4	Heavy damage	1.6	1.9	2.4	2,8	3,1
5	Malee Attack Range	20	24	27	30	33
6	Ranged Attack Range	18	23	26	29	32
7	Move Speed	12	16	20	25	29
8	Attack Cooldown	1,1	1	0.9	0.8	0,7

Table 14. Value's f the Enemies 2

Power Coefficients of enemies' features

The features of the enemies also have coefficients which indicate the effect of the particular features of the enemies.

Again, the values describe merely an average player's experience.

The product of health values and its coefficients will specify a value which should be surpassed by the total power of the spells.

The Sum product of the indices 2,3...8 will determine the damage of the enemy on the player.

Indices	Features	Power Coefficients
1	Current health	4
2	Damage	5
3	Light Damage	12
4	Heavy damage	8
5	Malee Attack Range	6
6	Ranged Attack Range	5
7	Move Speed	6
8	Attack Cooldown	-6

Table 15. Power Coefficient's of Enemies' Features

Sets

Letter	Definition	Indices
j	Stages of games (also levels of enemies)	1, 2, 3, 4, 5, 6, 7, 8, 9, 10
k	Features of enemies	1, 2, 3, 4, 5, 6, 7, 8
l	Name of the spells	a, b, c, d, e, f, g, h
s	Levels of spells	1, 2, 3
i	Features of spells	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11

Table 16. Sets

Parameters

- $v_{l,s,i}$: Values of spells
 c_i : Power Coefficients of spells' features
 $e_{j,k}$: Values of the enemies
 f_k : Power Coefficients of enemies' features

Decision Variables

- Z : the amount of coins when the player accomplished the last level of the game.
 $budget_j$: The budget at the beginning of the j th stage.

$points_j$: The total points which the player obtains when he/she passes the stage j. Its value is equal to the result of the formula: Player's damage on enemy – Enemy's damage to the player.

$spent_j$: The number of coins which are spent to buy spells at the beginning of the level j.

$gain_ratio$: The ratio, which indicates how much coins will be received according to the value of **$points_j$**

$units_{j,l,s}$: This variable will demonstrate the amount of “l” spells from “s” level which is purchased in the stage of j. These amounts will designate just an example of the combination of purchased spells and will be according to an average player.

$price_{l,s}$: This variable will indicate the prices of spells. These will be fixed numbers in the game and won't change according to the players.

Objective Function

$$z = budget_{10} - spent_{10} + (points_{10} * gain_ratio)$$

The value of “z” determines how many coins are left when the game finished successfully.

For a balanced gaming experience, this number should be minimal, since the player should be able to pass all the stages with minimal sources. In this way, a balanced gaming experience is provided that is neither too difficult nor too easy to finish it.

Constraints

$$budget_1 = 300$$

- ❖ The first amount of coins that the player has at the beginning of the game.

$$price_{a,1} = 4$$

- ❖ Price of the first level of spell “a” is 4.

$$\text{If } j < 2, units_{j,l,s} \leq 6$$

- ❖ The player can buy at most 6 times of each spell at the first stage.

$$\text{If } j \geq 2, \text{ } \mathbf{units}_{j,l,s} \leq 4$$

- ❖ The player can buy at most 4 times of each spell at the next stages.

$$\mathbf{spent}_j = \sum_{l=a}^h \sum_{s=1}^3 \mathbf{units}_{j,l,s} * \mathbf{price}_{l,s}$$

- ❖ The multiplication of the number of purchased items and their prices is equal to spent money at the level of j.

$$\mathbf{spent}_j \leq \mathbf{budget}_j$$

- ❖ Player cannot spend more money than he/she has.

$$\begin{aligned} \text{if } J > 1 & \Rightarrow \mathbf{spent}_j \geq \mathbf{spent}_{j-1} \\ \text{if } J > 1 & \Rightarrow \mathbf{budget}_j \geq \mathbf{budget}_{j-1} \end{aligned}$$

- ❖ The budget and the money spent shouldn't decrease as the player reaches the next level.

$$\text{if } J > 1 \Rightarrow \mathbf{budget}_j = \mathbf{budget}_{j-1} - \mathbf{spent}_{j-1} + (\mathbf{points}_{j-1} * \mathbf{gain_ratio})$$

- ❖ The remaining amount of coins in the j-1 level plus gained coins when the player passes the j-1 level build the budget of the next level.

$$\mathbf{price}_{l,s} \leq 500$$

- ❖ Prices of the spells cannot exceed 500 coins.

$$\text{if } s > 1 \Rightarrow 1 \leq \mathbf{price}_{l,s} - \mathbf{price}_{l,s-1} \leq 15$$

- ❖ The difference between the levels of the same level should be between 1 and 15.

$$\text{if } s > 2 \text{ and } l > 1 \Rightarrow 5 \leq \text{price}_{l,s-2} - \text{price}_{l-1,s} \leq 50$$

- ❖ The difference between the last level of a spell and the first level of the next spell should be between 5 and 50.
- ❖ These two constraints ensure the balance of pricing policy of the spells.

$$\text{if } j = 1 \text{ and } l > 2 \Rightarrow \text{units}_{j,l,s} = 0$$

$$\text{if } j = 2 \text{ and } l > 3 \Rightarrow \text{units}_{j,l,s} = 0$$

$$\text{if } j = 3 \text{ and } l > 4 \Rightarrow \text{units}_{j,l,s} = 0$$

$$\text{if } j = 4 \text{ and } l > 5 \Rightarrow \text{units}_{j,l,s} = 0$$

$$\text{if } j = 5 \text{ and } l > 6 \Rightarrow \text{units}_{j,l,s} = 0$$

$$\text{if } j = 6 \text{ and } l > 6 \Rightarrow \text{units}_{j,l,s} = 0$$

$$\text{if } j = 7 \text{ and } l > 6 \Rightarrow \text{units}_{j,l,s} = 0$$

$$\text{if } j = 8 \text{ and } l > 7 \Rightarrow \text{units}_{j,l,s} = 0$$

Some spells cannot be used in some certain levels. The visualization of these constraints is here :

stages	Spells							
<i>j</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>	<i>h</i>
1								
2								
3								
4								
5								
6								
7								
8								
9								
10								

Table 17. Stages of Spells

-The green cells show that the availability of the spells at certain stages.

$$\text{if } j = 9 \text{ and } l = 8 \Rightarrow \sum_{s=1}^3 \text{units}_{j,l,s} \geq 1$$

$$\text{if } j = 10 \text{ and } l = 8 \Rightarrow \sum_{s=1}^3 \text{units}_{j,l,s} \geq 1$$

- ❖ The last spell should be used in last stage of the game. These constraints ensure that no spell is redundant.

$$\sum_{l=a}^h \sum_{s=1}^3 \sum_{i=1}^{11} v_{l,s,i} * c_i * \text{units}_{j,l,s} \geq \sum_{j=1}^{10} e_{j,1} * f_1$$

- ❖ The damage of the player on the enemy should be greater than the total health (multiplied by the health coefficient of the enemy) of the enemy to pass the current j stage.

$$\text{points}_j = \sum_{l=a}^h \sum_{s=1}^3 \sum_{i=1}^{11} v_{l,s,i} * c_i * \text{units}_{j,l,s} - \sum_{k=1}^8 e_{j,k} * f_k + e_{j,1} * f_1$$

- ❖ The total points which the player obtains when he/she passes the stage j. Its value is equal to the result of the formula: Player's damage on enemy – Enemy's damage to the player. Since the first feature of the enemy is about health, it shouldn't be included in the damage function of the enemy.

3.4. Result

Prices of spells

Indices	Spells	Level 1	Level 2	Level 3
<i>a</i>	Untouchable Spell	4	5	6
<i>b</i>	Earth Wall Spell	11	12	13
<i>c</i>	Ground Spike Spell	29	43	58
<i>d</i>	Lightning Bolt Spell	108	123	138
<i>e</i>	Bow and Arrow Spell	188	203	218
<i>f</i>	Warding Shield Spell	268	283	298
<i>g</i>	Flame Ball Spell	348	363	378
<i>h</i>	Frost Spike Spell	428	443	458

Table 18. Prices of Spells

Units of spells

This is just an example of the combination of purchased spells which leads to success at the end of the game. The combination of spells can vary from player to player.

	Spells																							
	a			b			c			d			e			f			g			h		
Stages	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3
1	4	6	6	6	6	6																		
2	1		1				4	4																
3							2					2												
4												3												
5				4		4			4		4													
6				2					4		2	4												
7										4	1		2											
8	4					4			2	2	4					1								
9	3		3	1	1			1	2	1	1	1				1							1	
10	3	4	3	4	4	4	1	4	4	4	4					1						1		

Table 19. Units of Spells

Gain Ratio

$$= 0.8221$$

This is a fixed value in the game and won't change according to player.

Z

$$\begin{aligned} &= \text{budget}_{10} - \text{spent}_{10} + (\text{points}_{10} * \text{gain_ratio}) \\ &= 2598.81 - 2262 + (4190.60 * 0.822) \\ &= 3781.49 \end{aligned}$$

Again, this value can be different in different plays depending on the players. The main factor is the fact that the last amount of coins at the end of the game is minimum. The values of prices and gain_ratio which make the value of "z" minimum are the main goals in the model.

4. CONCLUSION

The incorporation of a mathematical modelling program such as GAMS (General Algebraic Modelling System) has effectively provided the means for achieving the optimal balance in the game. The model focused on the pricing policy of items in games and made sure that players get enough sources to be successful in the game. However, the Model does not provide superfluous number of sources to the players to make sure that players experience the consequences of their mistakes in the game. Through GAMS, it became possible to address and fine-tune multiples aspects of the game so that characters, accoutrements, skills and tactics are fairly balanced.

This kind of optimization has led to the creation of a fair and more entertaining gaming environment for users. Players now feel equal where one has to compete with others on the basis of skill, strategy and ideas and not with unequal advantages. This leads to a far more interesting and enjoyable gameplay, where the player is forced to think and approach a situation in a new way.

In conclusion, due to the application of GAMS for the optimization of in-game balance, this strategy has been deemed successful and efficient. They can make sure that players remain

interested and satisfied with the game, and immersed into the fair and balanced game environment. This is not only helps to increase the fun in the preliminary stages of the game but also guarantees its success in the future in the context of competitive gaming.

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